Plant foods and brain aging: a critical appraisal.

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In the 21st century, human aging will be one of the biggest challenges for most societies throughout the world. The decline in human fitness is a typical hallmark of the aging process. Aside from the cardiovascular system, the brain most often suffers significantly from the life-long impact of stressors, such as reactive oxygen and nitrogen species. Oxytosis, i.e. oxidative stress-induced cell death, has been identified to play a major role in the development and onset of chronic diseases. Foods, especially of plant origin, are rich in antioxidants and numerous in vivo data suggest that a diet rich in fruits and vegetables supports the maintenance of animal and human health. These beneficial effects also extend to the central nervous system, which, due to the presence of the blood-brain barrier, tightly controls the influx of metabolites and nutrients. In earlier studies the impact of antioxidant vitamins, such as alpha-tocopherol and ascorbic acid, on brain health has been of interest. Recently, the focus moved to assessing the potential of unsaturated fatty acids and secondary plant metabolites, particularly of polyphenols, to act as neuroprotectants. Considerable experimental evidence suggests that polyphenols and other plant-derived bioactivities affect animal and human brain function not only by directly lowering oxidative stress load but also by modulating various signal transduction pathways.